Tutorial 07

ENVX2001 – Applied Statistical Methods

Semester 1

Welcome

In this week's lecture we introduced you to Simple- and Multiple- Linear Regression. For both of these regression techniques we defined the key steps in model fitting and how to interpret model output.

You will learn how to:

1. Distinguish key similarities and differences between simple linear regression (SLR) and multiple linear regression (MLR) models
2. Implement the fitting process for each type of model
3. Test assumptions and evaluate model fit of SLR and MLR models.

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Simple linear regression vs Multiple Linear Regression

Table 1: Differences between MLR and SLR models

Element	Simple Linear Regression (SLR)	Multiple Linear Regression (MLR)
Number of predictors	1	2 or more
Model form	$Y = \beta_0 + \beta_1 x + \epsilon$	$Y = \beta_0 + \beta_1 x_1 + +\beta_2 x_2 + ... + \beta_n x_n + \epsilon$
How to fit in R	`lm(y ~ x, data = dataset)`	`lm(y ~ x1 + x2 + ... + xn, data = dataset)`
Assumptions	L.I.N.E.	L.I.N.E., collinearity
Model evaluation	r^2, residual plots	Adjusted r^2, residual plots, VIF
Conclusion	"as x increases, y also increases/decreases by β_1 units"	"as x_1 increases, y also increases/decreases by β_1 units, holding all other predictors constant"

Bird abundance

Fragmentation of forest habitat has an impact of wildlife abundance. This study looked at the relationship between bird abundance (bird ha^{-1}) and the characteristics of forest patches at 56 locations in SE Victoria.

The predictor variables are:

- `ALT` Altitude (m)
- `YR.ISOL` Year when the patch was isolated (years)
- `GRAZE` Grazing (coded 1-5 which is light to heavy)
- `AREA` Patch area (ha)
- `DIST` Distance to nearest patch (km)
- `LDIST` Distance to largest patch (km)

Import the data from the "Loyn" tab in the MS Excel file.

```
CODE
library(readxl)
loyn <- read_xlsx("data/mlr.xlsx", "Loyn")
```

Often, the first step in model development is to examine the data. This is a good way to get a feel for the data and to identify any issues that may need to be addressed. In this case, we will examine the data using histograms and a correlation matrix.

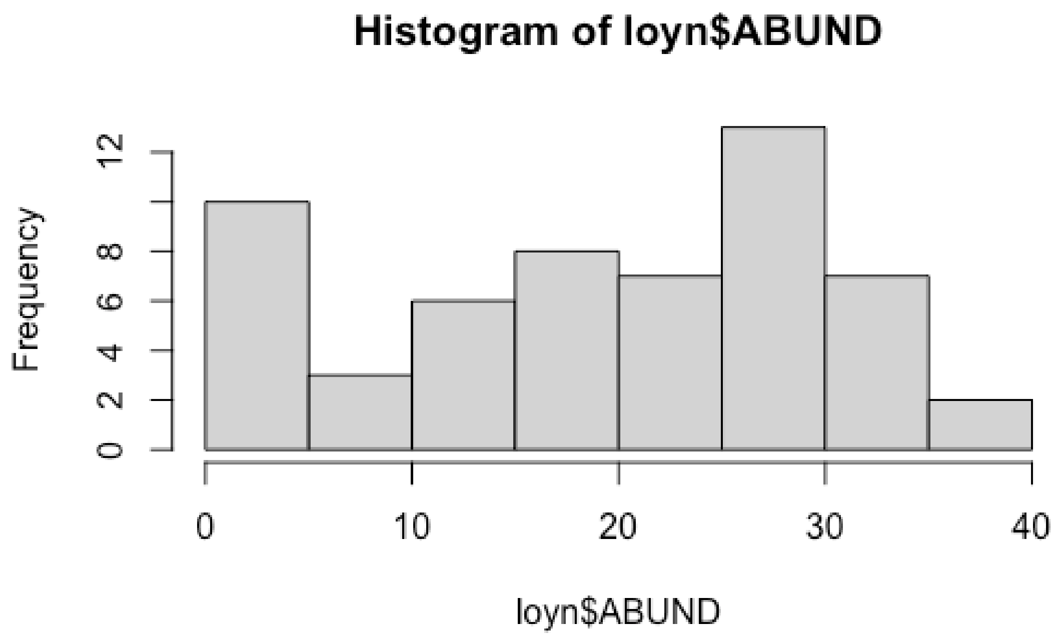
Histograms

There are a breadth of ways to create histograms in R. In each tab below you will find some different ways to create the same plot outputs.

`hist()`

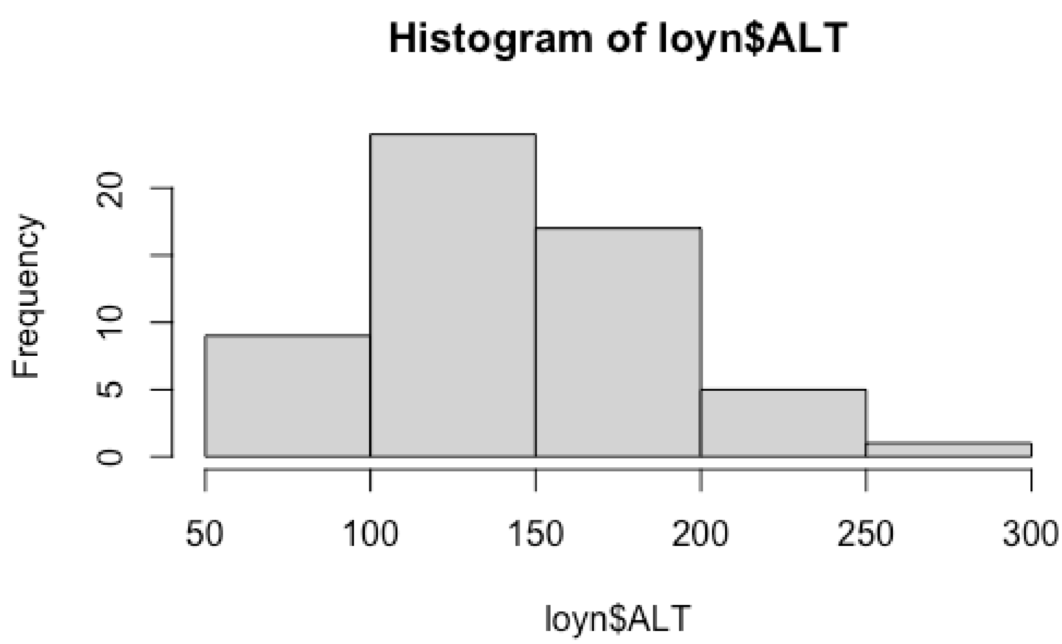
This is a straightforward way to create multiple histograms with `hist()`. The `par()` function is used to arrange the plots on the page. The `mfrow` argument specifies the number of rows and columns of plots.

```
CODE
# par(mfrow=c(3,3))
hist(loyn$ABUND)
```



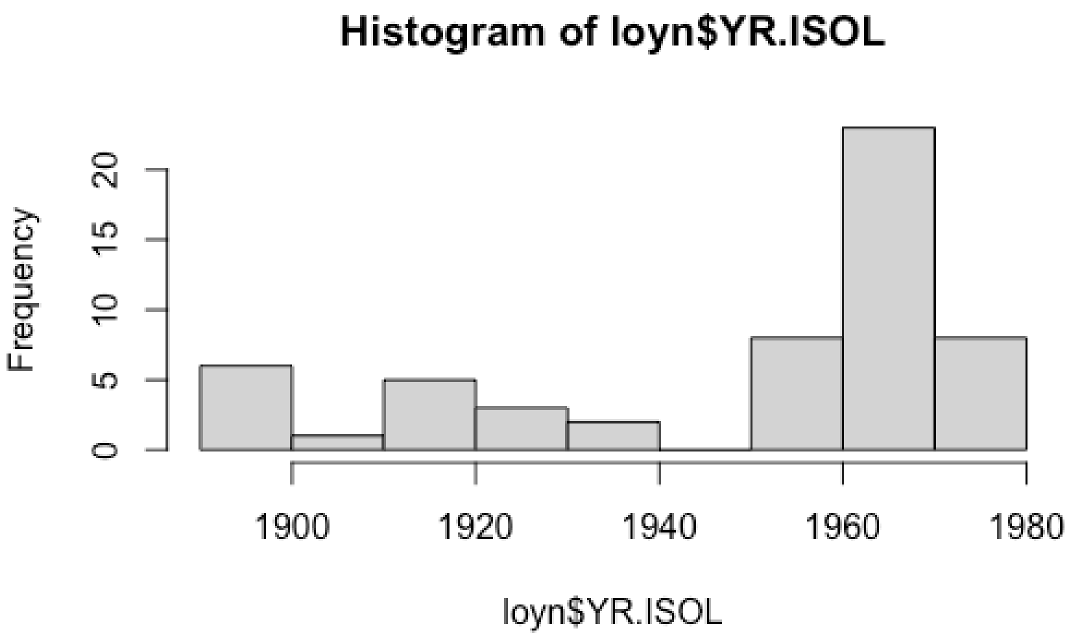
CODE

```
hist(loyn$ALT)
```



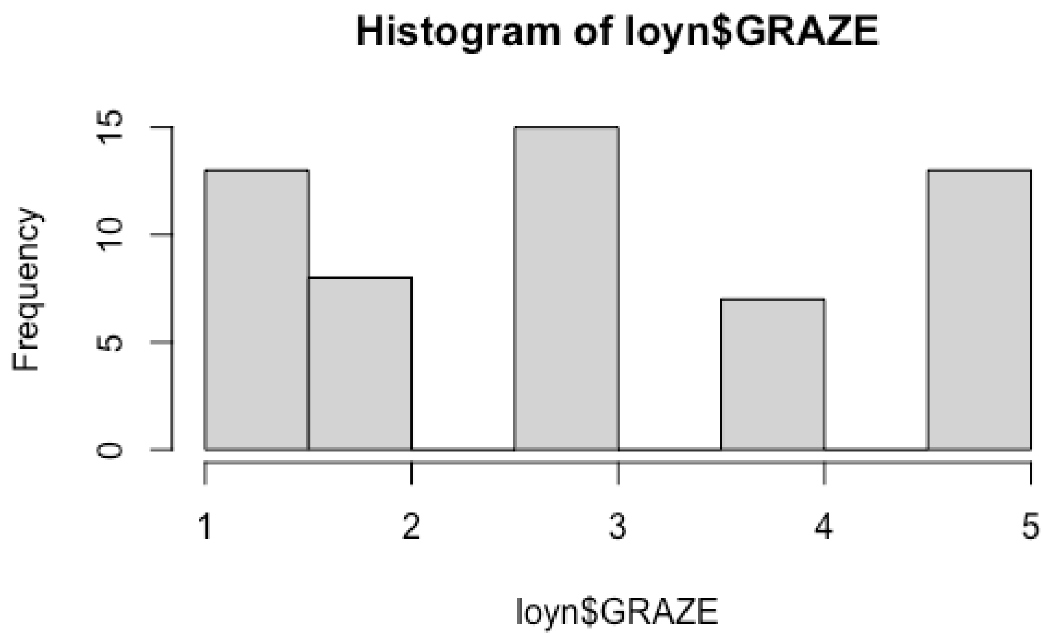
CODE

```
hist(loyn$YR.ISOL)
```



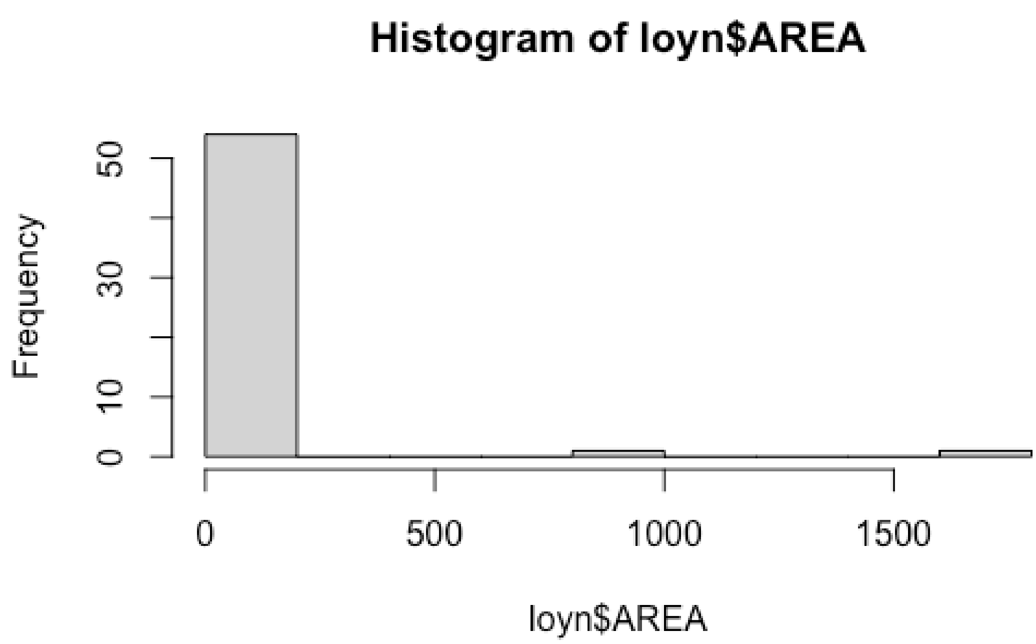
CODE

```
hist(loyn$GRAZE)
```



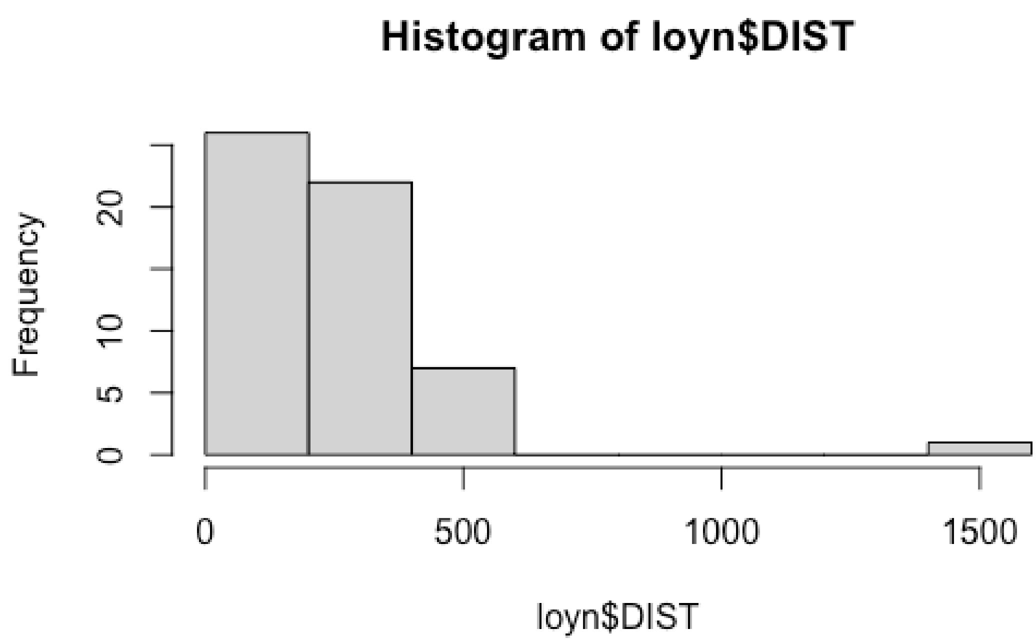
CODE

```
hist(loyn$AREA)
```



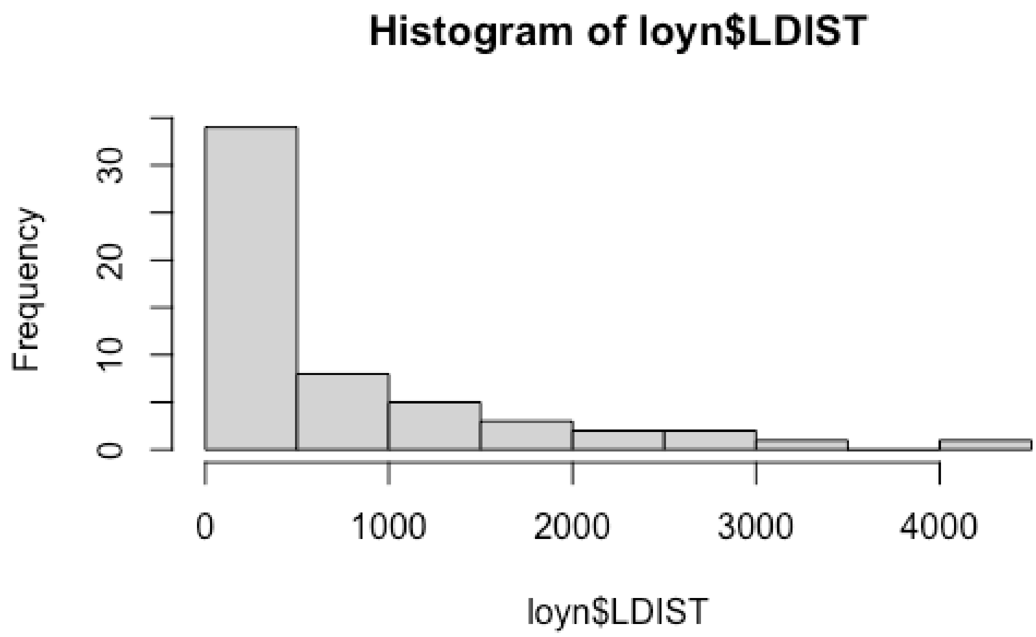
CODE

```
hist(loyn$DIST)
```



CODE

```
hist(loyn$LDIST)
```




```
CODE
# par(mfrow=c(1,1))
```

`hist.data.frame()` from `Hmisc`

The `Hmisc` package provides a function `hist.data.frame()` that can be used to create multiple histograms, which can be called by simply using `hist()`. You may need to tweak the `nclass` argument to get the desired number of bins, as the default may not look appropriate.

```
CODE
# install.packages("Hmisc")
library(Hmisc)
hist(loyn, nclass = 50)
```

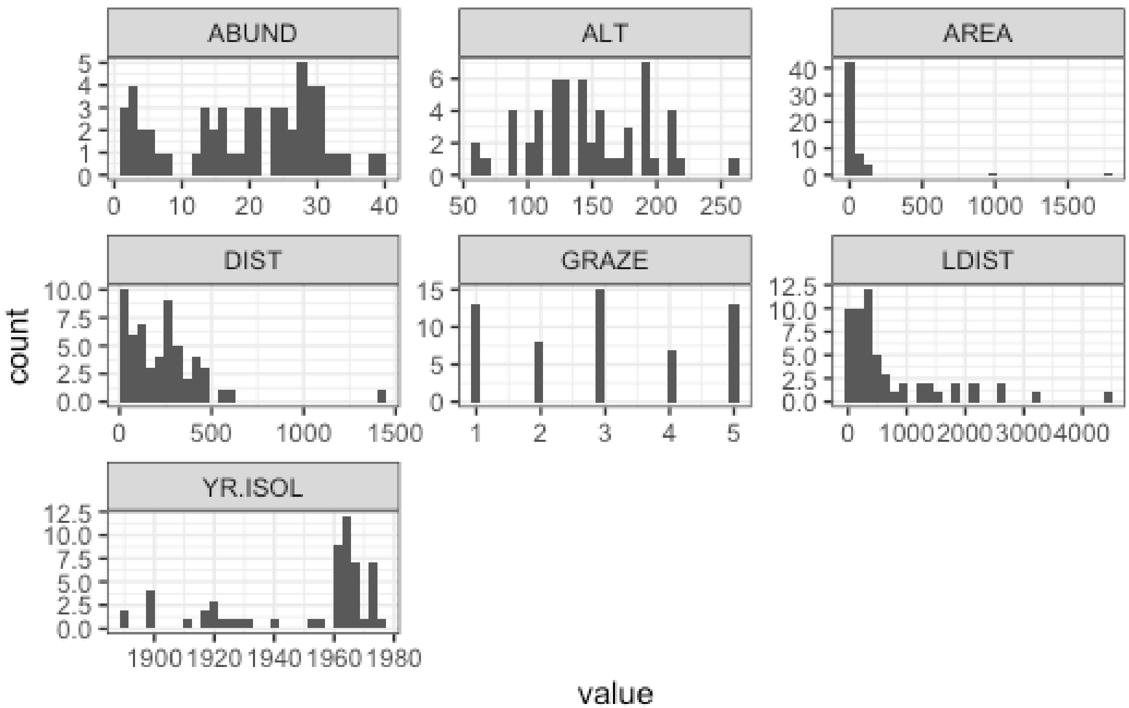
`ggplot()`

A more modern approach is to use `ggplot()` with `facet_wrap()` to arrange multiple plots on a single page. To do this, the `pivot_longer()` function from the `tidyr` package is used to reshape the data into a tidy format.

```
CODE
# tidy the data
loyn_tidy <- pivot_longer(loyn, cols = everything())

# plot
```

```
ggplot(loyn_tidy, aes(x = value)) +
    geom_histogram() +
    facet_wrap(~name, scales = "free") +
    theme_bw()
```

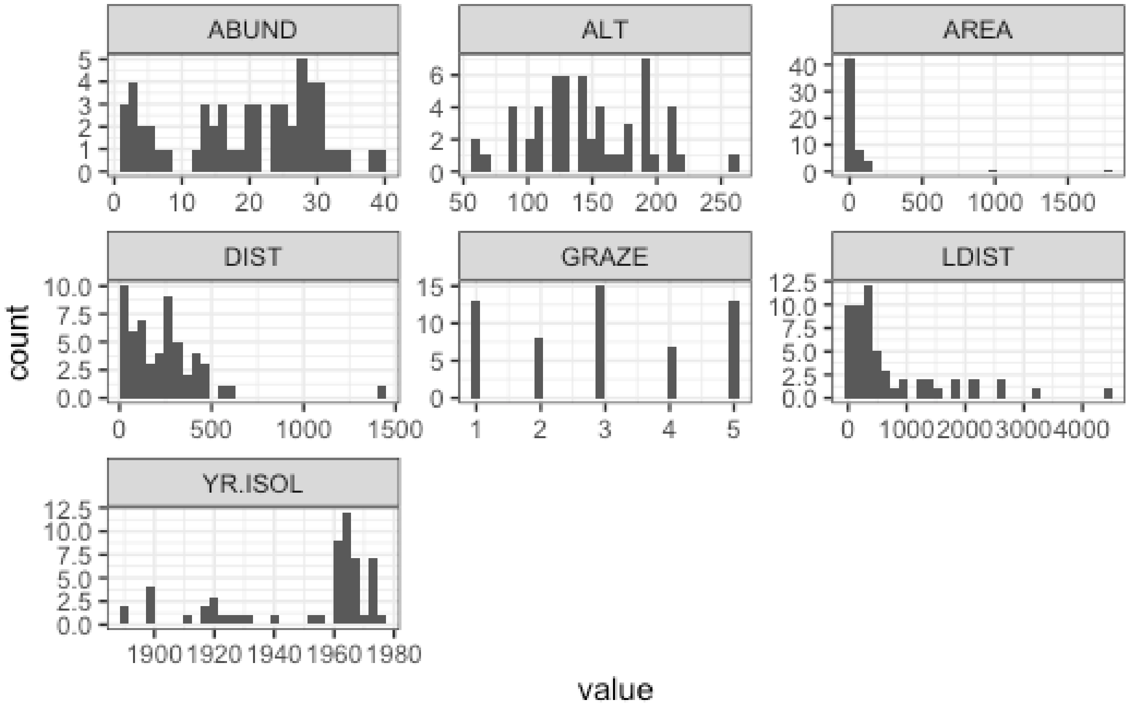


`ggplot()` with `dplyr`

Here we use the pipe operator `%>%` from `dplyr` to chain together a series of commands. The pipe operator takes the output of the command on the left and passes it to the command on the right (or below) the pipe. This means that we can create a series of commands that are executed in order.

CODE

```
loyn %>%
    pivot_longer(cols = everything()) %>%
    ggplot(aes(x = value)) +
    geom_histogram() +
    facet_wrap(~name, scales = "free") +
    theme_bw()
```



> **Question 1**
>
> Comment on the histograms in terms of leverage. *Hint: what is the relationship between leverage and skewness?*

Correlation matrix

Calculate the correlation matrix using `cor(Loyn)`.

CODE

```r
cor(loyn)
```

OUTPUT

```
             ABUND         AREA      YR.ISOL       DIST       LDIST
ABUND    1.00000000  0.255970206  0.503357741  0.2361125  0.08715258
AREA     0.25597021  1.000000000 -0.001494192  0.1083429  0.03458035
YR.ISOL  0.50335774 -0.001494192  1.000000000  0.1132175 -0.08331686
DIST     0.23611248  0.108342870  0.113217524  1.0000000  0.31717234
LDIST    0.08715258  0.034580346 -0.083316857  0.3171723  1.00000000
GRAZE   -0.68251138 -0.310402417 -0.635567104 -0.2558418 -0.02800944
ALT      0.38583617  0.387753885  0.232715406 -0.1101125 -0.30602220
             GRAZE        ALT
ABUND   -0.68251138  0.3858362
AREA    -0.31040242  0.3877539
YR.ISOL -0.63556710  0.2327154
DIST    -0.25584182 -0.1101125
LDIST   -0.02800944 -0.3060222
GRAZE    1.00000000 -0.4071671
ALT     -0.40716705  1.0000000
```

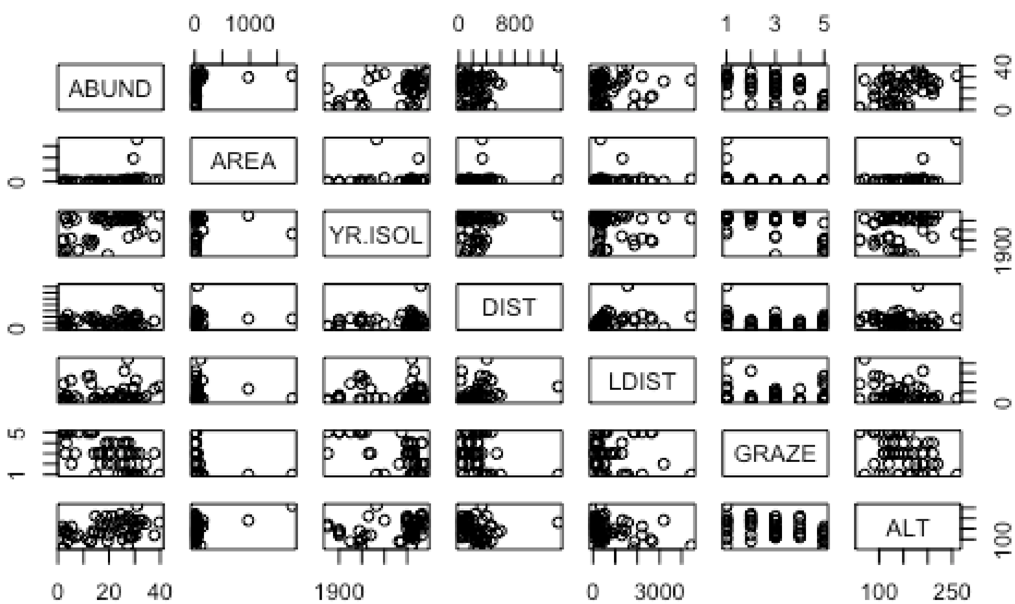
> **Question 2**
>
> Which independent variables are useful for predicting the dependent variable abundance? Is there evidence for multi-collinearity?

Plotting correlation

Examine correlations visually using `pairs()` or `corrplot()` from the `corrplot` package.

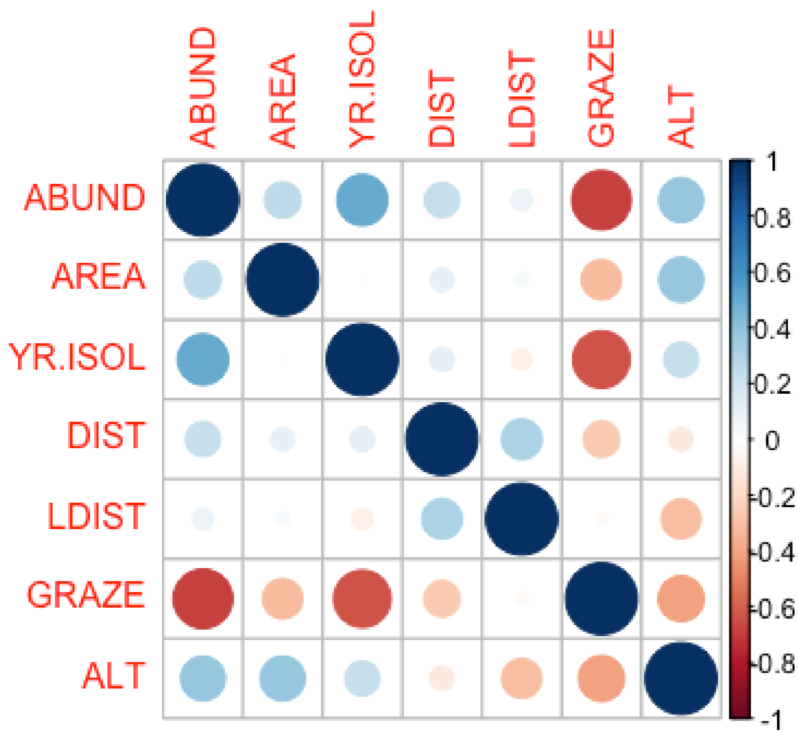
Scatterplot matrix

```
pairs(loyn)
```



Correlation matrix

```
library(corrplot)
corrplot(cor(loyn))
```



> **Question 3**
>
> Are there any trends visible from the plots?

> **Tip**
>
> We can also bring in variance inflation factors (VIF) to help us identify multi-collinearity, but that is done only after we have selected a model.

Transformations

The AREA predictor has a small number of observations with very large values. Apply a $\log_{10}$ transformation and label the new variable `Loyn$L10AREA`.

CODE
```
loyn$L10AREA <- log10(loyn$AREA)
```

> **Question 4**
>
> Why are we transforming AREA?

> **Question 5**
>
> Re-run `pairs(Loyn)` and create a histogram using the transformed value of AREA, how do the plots look?

```
CODE
hist(loyn$L10AREA)
pairs(loyn)
```

> **Question 6**
>
> In preparation for modelling, transform the remaining skewed variables, DIST and LDIST the same way you did for AREA and examine the histogram and pairs plots using these new variables.

Make sure you end up with two new variables labelled `loyn$L10DIST` and `loyn$L10LDIST`.

> **i Checkpoint**
>
> You should now be able to …[PUT THIS AFTER EACH OBJECTIVE]

Wrap-up

In this tutorial, **we explored factorial designs and their analysis using ANOVA. We learned about main effects, interactions, and blocking effects, and how to interpret ANOVA results in the context of factorial experiments. We also visualised these effects using ggplot2 and emmeans, enhancing our understanding of the relationships between factors and response variables in experimental designs.**

Example exam questions

1. In a 2-way factorial design with factors A (2 levels) and B (3 levels), describe how you would identify and interpret main effects and interactions using ANOVA. Provide an example of how to visualise these effects.

2. Explain the difference between a treatment design and an experimental design. Provide examples of each and discuss how they can be combined in a factorial design with blocking.

3. **If you conducted a factorial experiment with blocking and found a significant interaction between two factors, how would you interpret this result? Would you still consider the main effects of each factor? Justify your answer with reference to ANOVA principles.**